

SOLVED PRACTICE WORKBOOK

Geography 04 - Weathering, Mass Movement, Groundwater / India Erosion-Landslides-Groundwater - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

Geography 04 — Weathering, Mass Movement, Groundwater / India Erosion-Landslides-Groundwater - Learner-v2 Refreshed	
REFRESHED TEACHING NAVIGATION — BASIC/CORE FIRST	
01 THE EXOGENIC PROCESS SYSTEM: WEATHERING, EROSION, TRANSPORT, DEPOSITION, MASS	02 CONTROLS OF WEATHERING: ROCK, STRUCTURE, CLIMATE, RELIEF, ORGANISMS
03 MECHANICAL, CHEMICAL AND BIOLOGICAL WEATHERING	04 REGOLITH, PEDOGENESIS, SOIL HORIZONS AND BLACK SOIL
05 SOIL EROSION: SHEET, RILL, GULLY, RAVINE AND BOUNDED WIND/COASTAL LINKS	06 MASS-MOVEMENT TAXONOMY: CREEP, SOLIFLUCTION, FALLS, SLIDES, SLUMPS, FLOWS
07 SLOPE STABILITY: SHEAR STRESS, SHEAR STRENGTH, WATER AND TRIGGERS	08 INDIA LANDSLIDE GEOGRAPHY: HIMALAYA, NORTH-EAST, WESTERN GHATS AND NILGIRIS
09 LANDSLIDE INVENTORY, SUSCEPTIBILITY, FORECASTING LIMITS AND MITIGATION	10 SUBSIDENCE VERSUS LANDSLIDE: THE BOUNDED JOSHIMATH FRAME
11 GROUNDWATER IN THE HYDROLOGIC CYCLE AND THE WATER BUDGET	12 POROSITY, PERMEABILITY, INFILTRATION, PERCOLATION AND SUBSURFACE ZONES
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Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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Geography 04 - Weathering, Mass Movement, Groundwater / India Erosion-Landslides-Groundwater

BASIC MCQS / REMEDIATION

Practice rule

These 48 questions are original UPSC-style practice. The generation pipeline uses the stable geography -04 seed to balance and non-pattern correct-option positions while preserving the correct answer content. Questions 41-48 are remedial.

MCQ 1 - Exogenic boundary

Which statement correctly separates weathering from erosion?

A. Weathering changes rock in situ, whereas erosion includes detachment and transport by an external agent. B. Erosion occurs only after soil has formed. C. Denudation excludes mass wasting. D. Weathering always moves sediment downhill.

Answer: A.

Explanation: Weathering is defined by absence of transport; denudation is the wider lowering system.

MCQ 2 - Denudation system

Which sequence best represents a source-transfer-storage system?

A. Deposition -> uplift -> no further change B. Mass wasting -> weathering only after transport C. Weathering -> regolith -> entrainment/transport -> deposition -> possible remobilisation D. Soil formation -> no erosion

Answer: C.

Explanation: Weathered material can move, be stored and later be remobilised.

MCQ 3 - Weathering controls

Which is the most accurate statement?

A. Rock structure matters only in deserts. B. Time always creates deep soil even on rapidly eroding slopes. C. Climate alone fixes the same weathering product in every rock. D. Climate interacts with mineralogy, joints, relief, organisms, time and land use to control weathering.

Answer: D.

Explanation: Weathering is a multivariable process.

MCQ 4 - Thermal and unloading

Which distinction is correct?

A. Unloading requires salt crystallisation. B. Thermal expansion stresses rock through temperature cycling; unloading creates pressure-release sheet joints. C. Thermal expansion changes feldspar directly to clay. D. Both terms mean frost action.

Answer: B.

Explanation: The two mechanisms are mechanical but not identical.

MCQ 5 - Frost action

Frost weathering is strongest where:

- A. Only chemical solution occurs. B. Water repeatedly enters joints, freezes and thaws around the freezing point. C. The rock remains permanently dry. D. Temperatures never approach freezing.

Answer: B.

Explanation: Water access and repeated freeze-thaw cycles are essential.

MCQ 6 - Salt crystallisation

Salt weathering most directly involves:

- A. Crystal growth or hydration pressure within pores and cracks. B. A falling water table around a pumping well. C. Downslope sliding along a bedding plane. D. Carbonic acid dissolving limestone only.

Answer: A.

Explanation: Evaporation concentrates salts and growing crystals disrupt rock.

MCQ 7 - Wetting and drying

Repeated wetting and drying especially weakens:

- A. A confined aquifer by artesian flow. B. Freshwater head at a coast only. C. Clay-rich material that swells and shrinks. D. Only unjointed quartzite.

Answer: C.

Explanation: Moisture cycling creates fatigue in expansive materials.

MCQ 8 - Carbonation

Carbonation is best described as:

- A. Physical expansion during freezing. B. Carbon-dioxide-bearing water forming weak carbonic acid that dissolves carbonate minerals. C. Mechanical removal by wind. D. Oxygen reacting with iron only.

Answer: B.

Explanation: Carbonation is a chemical dissolution pathway central to limestone.

MCQ 9 - Hydrolysis

Hydrolysis can transform feldspar mainly into:

- A. Fresh volcanic glass without alteration. B. Clay minerals and dissolved ions. C. Pure metallic iron. D. Glacial till.

Answer: B.

Explanation: Water reacts with silicate minerals and creates secondary minerals.

MCQ 10 - Oxidation-reduction

Which statement is correct?

- A. Reduction is a mechanical fracture process. B. Redox has no relation to drainage. C. Iron-bearing minerals may oxidise in oxygenated conditions, while reducing conditions can change iron to more soluble forms. D. Oxidation requires no oxygen-bearing pathway.

Answer: C.

Explanation: Redox state depends strongly on oxygen and water conditions.

MCQ 11 - Biological weathering

Plant roots and microbes can:

- A. Operate only after erosion has ended.
- B. Create artesian pressure.
- C. Prevent all weathering beneath vegetation.
- D. Widen joints physically and accelerate chemical alteration through organic acids.

Answer: D.

Explanation: Biological action commonly reinforces physical and chemical processes.

MCQ 12 - Regolith and soil

Which statement is correct?

- A. Regolith is weathered mantle; soil requires biological activity and horizon development.
- B. All regolith is mature fertile soil.
- C. The B horizon is always unweathered bedrock.
- D. Soil forms without parent material, organisms or time.

Answer: A.

Explanation: Soil is an organised natural body, not merely broken rock.

MCQ 13 - Eluviation and illuviation

Which pairing is correct?

- A. Eluviation is removal from an upper horizon; illuviation is accumulation in a lower horizon.
- B. Both mean surface erosion.
- C. Illuviation occurs only in deserts.
- D. Eluviation means artesian flow.

Answer: A.

Explanation: The pair describes vertical translocation within a soil profile.

MCQ 14 - Black soil

Black cotton soil is most closely linked to:

- A. Recent marine coral deposits.
- B. Granite weathering alone in a perhumid climate.
- C. Weathering of Deccan basaltic/fissure-volcanic parent material and clay-rich shrink-swell behaviour.
- D. Unweathered shale with no clay.

Answer: C.

Explanation: Parent material and seasonal moisture help explain regur properties.

MCQ 15 - Sheet-rill-gully

Which order shows increasing concentration and incision by runoff?

- A. Rill -> ravine -> sheet -> gully
- B. Sheet -> rill -> gully -> ravine
- C. Ravine -> sheet -> rill -> gully
- D. Gully -> sheet -> ravine -> rill

Answer: B.

Explanation: Runoff becomes progressively channelised and erosive.

MCQ 16 - Wind erosion

Which transport sequence is associated with wind erosion?

- A. Creep, slump and artesian rise
- B. Surface creep, saltation and suspension
- C. Infiltration, percolation and recharge
- D. Solution, carbonation and hydrolysis

Answer: B.

Explanation: Wind moves coarse grains near the surface and finer grains in suspension.

MCQ 17 - Creep

A classic field indicator of soil creep is:

A. A single deep ocean trench. B. A flowing artesian well. C. Slowly tilted fences, posts or tree trunks on a slope. D. A fresh lava flow.

Answer: C.

Explanation: Creep is gradual and produces subtle deformation.

MCQ 18 - Solifluction

Solifluction refers to:

A. Dry rockfall from a vertical cliff only. B. Coastal saline upconing. C. Chemical oxidation of iron. D. Slow flow of water-saturated surface material over frozen or impermeable ground.

Answer: D.

Explanation: Saturation above a restricting layer promotes slow lobate movement.

MCQ 19 - Slump

A slump is distinguished by:

A. Rotational movement along a curved failure surface, often with back-tilted blocks. B. Uniform sheet wash. C. Free fall without a failure surface. D. Only wind transport.

Answer: A.

Explanation: Rotational geometry is the diagnostic clue.

MCQ 20 - Debris flow

Which description best fits a debris flow?

A. Groundwater moving through an aquitard. B. A coherent block moving on one plane only. C. Imperceptible soil creep. D. A rapid, water-rich, deforming mixture of coarse and fine material, often channelised.

Answer: D.

Explanation: Water and mixed debris produce flow-like motion.

MCQ 21 - Slope threshold

A slope becomes unstable when:

A. Driving shear stress exceeds available resisting shear strength. B. An inventory map is published. C. Porosity becomes zero everywhere. D. Rainfall begins regardless of material and slope.

Answer: A.

Explanation: Failure is a stress-strength threshold problem.

MCQ 22 - Pore-water pressure

Why can saturation reduce slope stability?

A. It converts every slide into subsidence. B. It always cements grains permanently. C. It may add weight and raise pore-water pressure, lowering effective resistance. D. It removes all joints.

Answer: C.

Explanation: Water can increase driving stress while reducing effective friction.

MCQ 23 - Toe undercutting

Toe erosion or excavation can destabilise a slope because it:

- A. Prevents runoff concentration. B. Always increases cohesion. C. Raises the potentiometric surface in all aquifers.
- D. Removes support from the lower slope and may steepen the profile.

Answer: D.

Explanation: Loss of toe support changes slope-force balance.

MCQ 24 - Himalaya-Ghats comparison

Which comparison is most defensible?

- A. The Himalaya combines young tectonic relief and fractured material; the Western Ghats often combine weathered escarpments, intense orographic rain and land-use/drainage change. B. Both regions have identical tectonic history.
- C. Western Ghats landslides are always earthquake-triggered. D. Himalayan landslides have no rainfall link.

Answer: A.

Explanation: The regions share triggers but differ in geological predisposition.

MCQ 25 - Inventory versus forecast

A landslide inventory primarily records:

- A. The exact date of every future failure. B. Only rainfall totals. C. Past mapped or reported events and their attributes.
- D. Groundwater quality samples.

Answer: C.

Explanation: Inventory supports pattern analysis but is not deterministic prediction.

MCQ 26 - Susceptibility map

A susceptibility map most directly shows:

- A. Guaranteed annual landslide counts. B. Only exposed population. C. Relative spatial propensity based on terrain and conditioning factors. D. The official compensation amount.

Answer: C.

Explanation: Susceptibility is a spatial screening product.

MCQ 27 - Early warning

Which statement about landslide early warning is correct?

- A. It removes weak rock. B. It predicts every failure with certainty. C. It replaces drainage and land-use planning. D. It can reduce exposure if thresholds, communication and action work, but it does not stabilise the slope.

Answer: D.

Explanation: Warning and source mitigation perform different functions.

MCQ 28 - Subsidence versus landslide

Which distinction is correct?

- A. Both are always identical. B. Subsidence is ground lowering/settlement; a landslide is downslope mass movement.
- C. Subsidence can occur only at coasts. D. A landslide has no shear movement.

Answer: B.

Explanation: They can coexist but should not be used as synonyms.

MCQ 29 - Joshimath evidence

The safest Joshimath conclusion is:

A. One visible crack proves one cause. B. All change is caused by groundwater pumping alone. C. Use a multi-causal frame involving ground material, drainage/water, slope and loading, with integrated evidence. D. Monitoring removes uncertainty.

Answer: C.

Explanation: The case requires bounded, evidence-led attribution.

MCQ 30 - Water budget

Which equation is conceptually correct?

A. Pumping creates recharge automatically. B. Discharge is not part of the aquifer budget. C. Storage always equals rainfall. D. Change in groundwater storage = recharge + inflow - discharge - pumping - outflow.

Answer: D.

Explanation: Groundwater storage changes through multiple inflows and outflows.

MCQ 31 - Infiltration-percolation

Which distinction is correct?

A. Both mean pumping. B. Infiltration is entry at the surface; percolation is subsequent downward movement through material. C. Percolation happens before surface entry. D. Infiltration always reaches the aquifer.

Answer: B.

Explanation: Surface entry does not guarantee deep recharge.

MCQ 32 - Porosity-permeability

Why can clay be porous but poorly permeable?

A. Permeability means total pore volume only. B. It contains no water. C. It may contain many tiny pores that are poorly connected for rapid flow. D. It is always crystalline limestone.

Answer: C.

Explanation: Storage space and connected transmission are different properties.

MCQ 33 - Aquifer-aquitard

Which pairing is correct?

A. Aquifer-zone of aeration only; aquitard-water table. B. Aquifer-surface river; aquitard-rainfall. C. Aquifer-completely impermeable; aquitard-unlimited flow. D. Aquifer-stores and transmits usable groundwater; aquitard-transmits much more slowly.

Answer: D.

Explanation: The distinction is relative hydraulic transmission.

MCQ 34 - Confined aquifer

A confined aquifer is:

A. Necessarily saline. B. A permeable unit bounded by lower-permeability layers and commonly under pressure. C. Unable to recharge anywhere. D. Always exposed directly at the local ground surface.

Answer: B.

Explanation: Confinement changes head conditions but does not create unlimited water.

MCQ 35 - Artesian flow

A well flows at the surface without pumping only when:

A. The potentiometric level of the confined aquifer lies above ground level. B. The well is merely deep. C. The aquifer is unconfined and dry. D. Clay has high permeability.

Answer: A.

Explanation: Artesian rise is pressure-controlled and flowing conditions are conditional.

MCQ 36 - Perched aquifer

A perched water body forms when:

A. The entire basin becomes confined. B. Seawater rises under every coastal well. C. A river erodes a ravine. D. A local low-permeability lens holds saturation above the regional water table.

Answer: D.

Explanation: Perched aquifers are local and commonly seasonal.

MCQ 37 - Hard-rock aquifer

Groundwater in crystalline hard-rock terrain is commonly stored and transmitted through:

A. Only primary intergranular pores like loose sand. B. Weathered mantles, joints, fractures and other secondary openings. C. An unlimited underground lake. D. Clay aquicludes alone.

Answer: B.

Explanation: Secondary porosity creates highly variable local yields.

MCQ 38 - Cone of depression

A cone of depression is:

A. The local lowering of hydraulic head around a pumping well. B. A gully head in a ravine. C. The saline interface at the coast only. D. A soil horizon.

Answer: A.

Explanation: Pumping produces spatial drawdown around the well.

MCQ 39 - Safe and sustainable yield

Which statement is most accurate?

A. Sustainable yield considers ecological, quality, social and long-term effects beyond a simple recharge-volume rule. B. Safe yield always equals annual recharge. C. Sustainable yield ignores baseflow and springs. D. Both terms mean unlimited pumping.

Answer: A.

Explanation: Groundwater sustainability is broader than arithmetic balance.

MCQ 40 - Coastal intrusion

Excessive coastal pumping can:

A. Push the saline interface seaward automatically. B. Lower freshwater head and induce landward interface movement or local upconing. C. Prove surface storm-surge flooding. D. Remove all salinity from soil.

Answer: B.

Explanation: Hydraulic-head decline is the central mechanism.

MCQ 41 - Remedial: weathering versus erosion

A rock cracks in place and is later carried away by a stream. Which classification is correct?

A. Cracking is weathering; stream removal is erosion and transportation. B. Both stages are deposition. C. Cracking is erosion because rock changed. D. Stream transport is weathering.

Answer: A.

Explanation: Separate in-situ breakdown from agent-driven removal.

MCQ 42 - Remedial: infiltration versus percolation

Rain enters the soil but is later stopped by a compact layer. What is certain?

A. Infiltration occurred, but deep percolation/recharge may have been limited. B. Recharge must equal rainfall. C. No infiltration occurred. D. The aquifer became artesian.

Answer: A.

Explanation: Entry and deep downward transfer are different.

MCQ 43 - Remedial: porosity versus permeability

Which material can have high porosity but low permeability?

A. Open well-sorted gravel only. B. A fully open fracture. C. A river channel. D. Clay with many small poorly connected pores.

Answer: D.

Explanation: Clay is the standard close-option trap.

MCQ 44 - Remedial: aquifer type

Water rises in a well above the top of the tapped aquifer but not above ground. This indicates:

A. A perched aquifer necessarily. B. No pressure. C. An unconfined water table at ground level. D. A confined artesian condition that is not flowing at the surface.

Answer: D.

Explanation: Artesian means pressure rise; flowing artesian is a narrower case.

MCQ 45 - Remedial: landslide type

A coherent mass moves downslope along a curved surface and leaves a head scarp. It is most likely:

A. Soil creep only. B. A rotational slide or slump. C. Saltation. D. A rockfall.

Answer: B.

Explanation: Curved failure geometry and back rotation distinguish a slump.

MCQ 46 - Remedial: rainfall causation

After intense rain a slope fails. Which conclusion is valid?

A. The event was predicted by any susceptibility map. B. Rain is necessarily the sole cause. C. Rain may be the trigger, while material, structure, drainage and cutting require investigation as predispositions. D. Road cutting can never matter.

Answer: C.

Explanation: Trigger and predisposition must be separated.

MCQ 47 - Remedial: quantity-quality

A falling water table proves:

A. Seawater intrusion in every inland basin. B. That recharge structures are effective. C. Arsenic contamination everywhere. D. Only a quantity trend at the measured location; contamination requires separate sampling and pathway evidence.

Answer: D.

Explanation: Quantity and quality evidence are distinct.

MCQ 48 - Remedial: recharge effectiveness

Which is the best evaluation of a recharge programme?

A. Assume every borewell is a recharge shaft. B. Ignore extraction changes. C. Check aquifer fit, source-water quality, maintenance, demand and measured level/quality trends. D. Count structures only.

Answer: C.

Explanation: Inputs must be separated from hydrological outcomes.

PYQS AND ANSWER PRACTICE

PYQ control note

- Exact wording below was checked against the routed local official papers.
- **Official Set-A keys:** 2024 Q7 = A; 2025 Q28 = C.
- **Key unavailable locally:** 2020 Q98, 2021 Q64 and 2023 Q76. Any independent solution is explicitly non-official.
- Mains has no UPSC model key; the answers below are independent examiner-grade models.
- Direct owner is shown first. Cross-owned questions are included because Topic 04 supplies essential physical-geography mechanics, but ownership is not transferred.

VERIFIED PYQ 1 - UPSC Prelims 2021 GS-I Q64: black cotton soil

Exact question: The black cotton soil of India has been formed due to the weathering of

A. brown forest soil B. fissure volcanic rock C. granite and schist D. shale and limestone

Official-key status: No official local key is available.

Independent solution - NOT AN OFFICIAL KEY: B. The Deccan black-soil region is associated with weathering of basaltic lava flows, described in the option as fissure volcanic rock. Brown forest soil is not parent rock; granite/schist and shale/limestone do not explain the classic regur association.

VERIFIED PYQ 2 - UPSC Prelims 2023 GS-I Q76: groundwater withdrawal

Exact question:

Statement-I: According to the United Nations 'World Water Development Report, 2022', India extracts more than a quarter of the world's groundwater withdrawal each year.

Statement-II: India needs to extract more than a quarter of the world's groundwater each year to satisfy the drinking water and sanitation needs of almost 18% of world's population living in its territory.

Which one of the following is correct in respect of the above statements?

A. Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I B. Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I C. Statement-I is correct but Statement-II is incorrect D. Statement-I is incorrect but Statement-II is correct

Official-key status: No official local key is available.

Independent solution - NOT AN OFFICIAL KEY: C. Statement-I is consistent with the cited report framing. Statement-II is not a valid explanation because India's groundwater withdrawal is dominated by irrigation rather than a

necessary drinking-water-and-sanitation requirement proportional to population. The 10 August 2026 Ministry reply separately reports 87% irrigation, 11% domestic and 2% industrial shares for the 2025 assessment.

VERIFIED PYQ 3 - UPSC Prelims 2024 GS-I Q7: rainfall and weathering

Exact question:

Statement-I: Rainfall is one of the reasons for weathering of rocks. Statement-II: Rain water contains carbon dioxide in solution. Statement-III: Rain water contains atmospheric oxygen.

Which one of the following is correct in respect of the above statements?

A. Both Statement-II and Statement-III are correct and both of them explain Statement-I B. Both Statement-II and Statement-III are correct, but only one of them explains Statement-I C. Only one of the Statements II and III is correct and that explains Statement-I D. Neither Statement-II nor Statement-III is correct

Official local Set-A key: A. Dissolved carbon dioxide enables weak carbonic-acid/carbonation reactions; dissolved oxygen supports oxidation. Both can make rainfall a weathering agent.

VERIFIED PYQ 4 - UPSC Prelims 2025 GS-I Q28: chalk and clay

Exact question:

Statement-I: In the context of effect of water on rocks, chalk is known as a very permeable rock whereas clay is known as quite an impermeable or least permeable rock. Statement-II: Chalk is porous and hence can absorb water. Statement-III: Clay is not at all porous.

Which one of the following is correct in respect of the above statements?

A. Both Statement-II and Statement-III are correct and both of them explain Statement-I B. Both Statement-II and Statement-III are correct but only one of them explains Statement-I C. Only one of the Statements II and III is correct and that explains Statement-I D. Neither Statement-II nor Statement-III is correct

Official local Set-A key: C. Statement-II is correct and explains chalk's ability to admit water; Statement-III is false because clay can be porous but has very low permeability due to tiny, poorly connected pores.

SUPPLEMENTARY CROSS-OWNED PYQ 5 - UPSC Prelims 2020 GS-I Q98: CGWA

Exact question:

Consider the following statements:

1. 36% of India's districts are classified as "overexploited" or "critical" by the Central Ground Water Authority (CGWA).
2. CGWA was formed under the Environment (Protection) Act.
3. India has the largest area under groundwater irrigation in the world.

Which of the statements given above is/are correct?

A. 1 only B. 2 and 3 only C. 2 only D. 1 and 3 only

Official-key status: No official local key is available. The workbook does not convert an external or inferred key into an official fact. The concept route is institutional: CGWA was constituted under the Environment (Protection) Act framework; dated percentages must be treated cautiously.

VERIFIED PYQ 6 - UPSC Mains 2018 GS-I Q14, 15 marks

Exact question: "The ideal solution of depleting ground water resources in India is water harvesting system." How can it be made effective in urban areas? (Answer in 250 words)

Model answer

Urban water harvesting can relieve groundwater stress only when it is designed as a water-budget and aquifer-management system rather than as a construction target.

First, cities should capture roof runoff for direct non-potable storage and route only suitably treated surplus to recharge. First-flush diversion, filtration and separation from sewage are essential because contaminated recharge

can damage an aquifer. Second, recharge structures must follow hydrogeological mapping: permeable recharge zones, depth to water, confining layers and contamination risk decide whether a pit, shaft, trench, well or restored tank is suitable.

Third, urban planning must preserve lakes, wetlands, floodplains and open recharge areas while reducing impervious cover through blue-green infrastructure. Storm-water drains should be integrated with detention and recharge rather than used merely to export runoff. Fourth, demand must fall through leakage control, reuse of treated wastewater, metering/regulation of bulk extraction and water-efficient fixtures; otherwise added recharge is overwhelmed.

Finally, municipal bodies need enforceable building provisions, maintenance registers, public aquifer/water-level maps and quality monitoring. The 3-10 August 2026 Ministry of Jal Shakti replies confirm that CGWB/NAQUIM, digital monitoring and recharge campaigns provide planning instruments, but structure counts do not prove recovery.

Conclusion: harvesting is effective when **capture + quality + aquifer fit + demand control + maintenance + measured outcomes** operate together.

Why this earns marks: It answers “how effective”, uses an urban-specific mechanism, names current institutions and adds an outcome-based qualification.

SUPPLEMENTARY CROSS-OWNED PYQ 7 - UPSC Mains GS-III 2019 Q18, 15 marks

Exact question: Disaster preparedness is the first step in any disaster management process. Explain how hazard zonation mapping will help disaster mitigation in the case of landslides. (Answer in 250 words)

Model answer

Landslide hazard zonation maps classify terrain by relative likelihood or intensity of slope failure using geology, slope, drainage, land cover, geomorphology and past-event inventory. They support mitigation before a trigger becomes a disaster.

At planning stage, the maps can keep settlements, hospitals, schools and lifeline corridors away from very-high-susceptibility slopes; guide alignment and design of roads, tunnels and utilities; and identify where site investigation must precede construction. At preparedness stage, zonation helps locate monitoring instruments, rainfall thresholds, evacuation routes, shelters, equipment and maintenance priorities. At mitigation stage, it supports targeted drainage, toe protection, retaining/anchoring, bioengineering and controlled spoil disposal rather than uniform treatment.

GSI's national inventory and susceptibility work provides the spatial base. Its June 2026 Chanmari West assessment illustrates why maps must be followed by field examination: a mapped high-susceptibility slope still required joint, toe-erosion, runoff and settlement-drainage diagnosis.

However, susceptibility is not the same as a date-specific forecast, and hazard is not risk. Exposure, vulnerability, map scale, changing land use, rainfall uncertainty and last-mile communication determine outcomes.

Conclusion: zonation is most effective when legally linked to land-use control, engineering design, maintenance, forecast dissemination and community drills.

Why this earns marks: It explains the decision uses of zonation and qualifies scale, timing and risk limitations.

SUPPLEMENTARY CROSS-OWNED PYQ 8 - UPSC Mains GS-I 2021 Q4, 10 marks

Exact question: Differentiate the causes of landslides in the Himalayan region and Western Ghats. (Answer in 150 words)

Model answer

Both regions experience gravity-driven failure, but their predispositions differ.

Himalayan region	Western Ghats
Young collision belt, active uplift and seismicity	Older escarpment/plateau-margin terrain
Very high relief, deep river incision and fractured/folded rocks	Deep weathering, lateritic/altere mantle and steep scarps
Snowmelt, monsoon saturation and earthquakes can trigger failure	Intense orographic monsoon rainfall and short saturated slopes dominate many events

Himalayan region	Western Ghats
Road cutting, tunnelling, toe erosion and unstable debris amplify risk	Quarrying, road/house cuts, plantation/land-use change and blocked drainage amplify risk

In both, rainfall is often a trigger rather than the entire cause; lithology, joints, slope geometry, drainage and human loading need site evidence.

Conclusion: mitigation must therefore be region-specific-tectonic and corridor-sensitive in the Himalaya, and drainage/weathered-mantle/land-use sensitive in the Ghats.

Why this earns marks: It obeys “differentiate”, gives paired causal categories and retains a shared multi-factor qualification.

SUPPLEMENTARY CROSS-OWNED PYQ 9 - UPSC Mains GS-III 2021 Q18, 15 marks

Exact question: Describe the various causes and the effects of landslides. Mention the important components of the National Landslide Risk Management Strategy. (Answer in 250 words)

Model answer

Landslides are gravity-driven movements of rock, debris or earth when driving stress exceeds resisting strength.

Causes: steep or undercut slopes; weak, weathered or jointed lithology; adverse bedding; old debris; saturation and pore-pressure rise during intense/prolonged rain; river/toe erosion; earthquake shaking; and human triggers such as road cutting, quarrying, loading, leakage, deforestation and unplanned drainage.

Effects: deaths and injuries; burial of settlements; road, rail, pipeline and power disruption; river blockage and flash-flood risk; sedimentation; loss of soil and forests; isolation of mountain communities; and recurring livelihood/fiscal costs.

Risk-management components: inventory and susceptibility/hazard zonation; site investigation and monitoring; rainfall/threshold-based forecasting with last-mile warning; land-use regulation and safer corridor alignment; drainage and slope treatment; bioengineering and designed retaining/toe works; enforcement of hill construction standards; community awareness, evacuation routes and drills; emergency access and post-event learning.

The GSI/NLFC ecosystem supplies inventory, susceptibility and forecast inputs, while NDMA/state authorities own broader disaster-management coordination. Forecasts reduce exposure but do not stabilise a slope.

Conclusion: an effective strategy connects **map -> regulate -> drain/stabilise -> warn -> evacuate -> update inventory**.

Why this earns marks: It covers causes, effects and strategy as three explicit demands and respects institutional ownership.

SUPPLEMENTARY CROSS-OWNED PYQ 10 - UPSC Mains GS-I 2024 Q14, 15 marks

Exact question: The groundwater potential of the Gangetic valley is on a serious decline. How may it affect the food security of India? (Answer in 250 words)

Model answer

The Gangetic valley's alluvial aquifers support a major irrigated foodgrain region. Persistent withdrawal beyond replenishment can therefore transmit hydrological stress into food insecurity.

falling head/yield -> deeper pumping + higher energy cost
-> delayed/less reliable irrigation
-> crop-yield and income risk
-> food availability, access and stability stress

Small and marginal farmers have less capacity to deepen wells, purchase pumps or absorb energy costs, so the first effect is often unequal access to irrigation. Rice-wheat and other water-intensive rotations can face timing and yield risk, while reduced dry-season buffer increases dependence on variable rainfall. Higher production costs can affect prices and procurement; quality deterioration can also make nominal groundwater unusable.

Yet tube-well irrigation has supported productivity and drought resilience. The solution is not indiscriminate pumping prohibition. It is conjunctive surface-groundwater use, crop diversification aligned with agro-climate, efficient irrigation,

rational energy/procurement signals, protection of recharge zones and wetlands, managed recharge where suitable, and aquifer-level community water budgets.

Conclusion: food security requires managing groundwater as production infrastructure and a common-pool resource, with equity for smallholders.

Why this earns marks: It links groundwater to all food-security pillars, explains the transmission chain and gives a balanced demand-side solution.

SUPPLEMENTARY CROSS-OWNED PYQ 11 - UPSC Mains GS-III 2025 Q8, 10 marks

Exact question: Seawater intrusion in the coastal aquifers is a major concern in India. What are the causes of seawater intrusion and the remedial measures to combat this hazard? (Answer in 150 words)

Model answer

Seawater intrusion is landward or upward movement of saline water into a coastal freshwater aquifer when freshwater hydraulic head becomes inadequate.

Causes: excessive pumping and clustered deep wells; reduced recharge through surface sealing or drought; coastal-aquifer geometry and preferential pathways; irrigation return/saline canals; and sea-level or storm stress where locally relevant. A pumping well may also draw saline water upward as **upconing**.

Measures: monitor groundwater head and salinity; cap, meter or redistribute abstraction; protect recharge zones; use suitably treated managed recharge; space and redesign wells; adopt efficient irrigation and less water-intensive uses; use site-specific subsurface/hydraulic barriers where feasible; and prevent pollution that compounds salinity.

Desalination may supply water but does not by itself restore aquifer pressure. Recharge without source-water and aquifer safeguards can worsen quality.

Conclusion: coastal aquifer security needs a monitored balance of pumping, recharge and land-use suited to local hydrogeology.

Why this earns marks: It begins with the hydraulic mechanism and matches each remedy to a cause.

SUPPLEMENTARY CROSS-OWNED PYQ 12 - UPSC Mains GS-III 2025 Q13, 15 marks

Exact question: Examine the factors responsible for depleting groundwater in India. What are the steps taken by the government to mitigate such depletion of groundwater? (Answer in 250 words)

Model answer

Groundwater depletes when withdrawal persistently exceeds recharge at the relevant aquifer scale.

Factors: irrigation-dominated extraction; water-intensive crops and procurement incentives; low-cost/unmetered farm energy; numerous private wells; urban and industrial demand; leakage and sealed recharge surfaces; drought and rainfall variability; loss of tanks/wetlands; and limited storage in weathered/fractured hard-rock aquifers. Weak aquifer-scale regulation turns an individually rational pump into a collective decline.

Government steps: CGWB conducts annual dynamic resource assessment, level/quality monitoring and technical studies. NAQUIM 1.0 mapped about 25 lakh sq km; NAQUIM 2.0 targets detailed priority-area studies. The Atal Bhujal pilot used community water-security planning in 8,203 Gram Panchayats across seven states. JSA: Catch the Rain, JSJB: CTR 2026, the Master Plan for Artificial Recharge, PMKSY components, water-body restoration and Jal Shakti Kendras support recharge and conservation. CGWA/state rules regulate specified extraction.

The 6-10 August 2026 Ministry replies report national recharge/extraction improvement and large structure counts, but these are not proof of recovery in every aquifer. Crop/energy incentives, metering, reuse, maintenance, quality safeguards and transparent local outcomes remain necessary.

Conclusion: scientific mapping and recharge must be coupled with demand reform and accountable state/local governance.

Why this earns marks: It examines natural, economic and institutional drivers and evaluates government measures rather than listing schemes.

ORIGINAL 10-MARK QUESTION 1 - process distinction

Question: Distinguish weathering, erosion and mass movement. Show how their interaction can create risk along a Himalayan road corridor. (150 words)

Model answer: Weathering breaks or alters rock in situ; erosion detaches and transports material through water, wind, ice or waves; mass movement transfers rock/debris downslope mainly under gravity. In a Himalayan road corridor, fractured rock and weathered regolith provide susceptible material. A road cut may steepen or undercut the slope and redirect drainage. Intense rain can infiltrate joints, add weight and raise pore-water pressure until resisting strength is exceeded. The resulting slide blocks the road; runoff may then entrain debris and erode the toe. GSI inventory/susceptibility work shows why geology, slope, rainfall and exposure must be combined. Yet neither road construction nor rainfall alone proves cause at a particular site. Site investigation, drainage design, toe support, slope treatment, spoil control, maintenance and warning/closure protocols are needed. **Why this earns marks:** definitions are exact, the causal chain is applied to India, and attribution is qualified.

ORIGINAL 10-MARK QUESTION 2 - porosity and permeability

Question: Explain why porosity alone cannot determine whether a rock or sediment is a productive aquifer. (150 words)

Model answer: Porosity measures the fraction of void space, whereas permeability measures whether pores or fractures are connected well enough to transmit water. Clay can contain abundant microscopic pore space yet release and transmit water very slowly; it can act as an aquitard. Well-sorted sand and gravel combine useful porosity with connected pores. Chalk may absorb and transmit water through matrix pores and fractures. Crystalline hard rock has little primary porosity but can yield groundwater through weathered mantles, joints and fractures. Productivity also depends on saturated thickness, recharge, boundary conditions and well design. Therefore aquifer assessment requires both storage and transmissivity evidence. **Why this earns marks:** it uses the exact close-option distinction, gives contrasting materials and connects the concept to India's hard-rock aquifers.

ORIGINAL 15-MARK QUESTION 1 - watershed management

Question: Examine why soil erosion and slope failure should be managed at watershed scale in India. Illustrate with distinct physiographic settings. (250 words)

Model answer: Soil erosion and slope failure are linked watershed processes because vegetation, slope length, drainage and infiltration control runoff energy, sediment transfer and pore-water conditions. In the Chambal tract, diffuse sheet wash can concentrate into rills, gullies and ravines; cover, contouring, gully-head control and catchment drainage must act upslope. Himalayan corridors combine high relief, fractured/weathered material, river undercutting, road cuts and intense rainfall; drainage, toe protection, designed stabilisation and hazard-sensitive alignment must work together. Western Ghats/Nilgiris slopes have a different mix of deep weathering/laterite, escarpment relief, orographic rain, quarry/road cuts and altered drainage. Catchment treatment can also improve infiltration and spring/aquifer recharge, but poorly sited sediment-laden structures may clog or destabilise slopes. Watershed scale does not eliminate plot-level tenure, maintenance and incentive problems. India therefore needs nested plot-slope-micro-watershed-aquifer planning supported by monitoring and community institutions. **Why this earns marks:** it examines a causal proposition, compares regions and matches interventions to mechanisms.

ORIGINAL 15-MARK QUESTION 2 - urban subsidence

Question: Analyse the relationship between groundwater extraction, geological conditions and land subsidence in Indian cities and plains. Suggest a risk-reduction framework. (250 words)

Model answer: Groundwater extraction can cause land subsidence where pumping lowers pore pressure in compressible aquifer-system sediments, transfers load to the grain skeleton and produces compaction. The effect is strongest where thick fine-grained layers are susceptible, withdrawal is concentrated and recharge is inadequate. Hard-rock cities may show a different response because storage is fracture-controlled; not every falling water table produces measurable subsidence. Urban loading, excavation, drainage leakage, cavities and natural consolidation can also contribute, so attribution requires geodetic, borehole, pumping and hydrogeological evidence. Impacts include differential settlement, cracked buildings, damaged drains and increased flood exposure. Risk reduction requires aquifer-wise extraction budgets; well registration and metering for major users; conjunctive supply and reuse;

recharge-zone protection and safe managed recharge; InSAR/GNSS and groundwater-level monitoring; geotechnical zoning; building/drainage audits; and public disclosure. **Conclusion:** subsidence policy must connect water management with geotechnical land-use planning while avoiding single-cause claims. **Why this earns marks:** it explains the compaction mechanism, distinguishes settings and proposes measurable interventions.

ORIGINAL 20-MARK QUESTION 1 - integrated landslide governance

Question: "A landslide disaster is produced as much by exposure and vulnerability as by slope failure." Analyse for India and propose an integrated strategy. (250 words)

Model answer: A landslide hazard is the physical probability and intensity of slope movement; disaster risk emerges when people and assets are exposed and vulnerable, while capacity is weak. India's Himalayan and North-Eastern belts combine high relief, fractured or weathered materials, seismicity, river incision and monsoon triggers. Western Ghats/Nilgiris commonly add deep weathering, orographic rain, slope cutting, quarrying and drainage disruption. Yet the same physical event produces different losses depending on settlement location, road redundancy, construction quality, poverty and warning access. GSI inventory and susceptibility mapping help avoid high-risk siting; the June 2026 Chanmari report shows why field diagnosis of joints, toe erosion, runoff and settlement drainage is still necessary. An integrated strategy should: regulate land use and carrying capacity; require site investigation and safe road/slope design; maintain catch drains and culverts; combine engineered stabilisation with bioengineering; monitor rainfall and movement; link forecasts to local communication, closures and evacuation; train communities; protect alternate access; and update inventories after events. Forecasts cannot stabilise slopes, and retaining walls cannot compensate for unmanaged water. **Conclusion:** India must join source treatment with exposure reduction and social preparedness. **Why this earns marks:** it uses the risk equation, regional differentiation, current GSI evidence and a layered strategy.

ORIGINAL 20-MARK QUESTION 2 - groundwater common-pool resource

Question: "Groundwater is a stock governed by flows, quality and institutions-not an invisible reserve waiting to be pumped." Analyse this proposition for India, including food security and coastal aquifers. (250 words)

Model answer: Groundwater storage changes through recharge, inflow, natural discharge, baseflow, pumping and inter-aquifer leakage; usable supply also depends on permeability and quality. The 2025 assessment reported in August 2026 distinguishes 448.52 BCM recharge, 407.75 BCM extractable resource and 247.22 BCM extraction. These national values cannot certify a local aquifer. In the Gangetic food belt, falling head raises lift and energy costs, reduces irrigation reliability and affects output, smallholder income and food-security stability. In hard-rock India, limited discontinuous weathered/fracture storage creates sharp local variability. At the coast, excessive pumping lowers freshwater head, enabling landward saline-interface movement and upconing. Arsenic, fluoride, nitrate and salinity further reduce usable stock through different pathways. CGWB assessment/monitoring, NAQUIM, Atal Bhujal learning and recharge programmes provide tools, but sustainable management also needs crop-energy reform, efficient irrigation, conjunctive use, reuse, quality safeguards, state/local regulation and aquifer-level participation. **Conclusion:** recharge is necessary but cannot substitute for demand management and accountable institutions. **Why this earns marks:** it integrates stock-flow, regional hydrogeology, quality, food and governance with current evidence.

DIAGRAM AND MAP PRACTICE

Diagram 1 - weathering to storage

ROCK -> weathering -> regolith -> [gravity movement / agent transport]
-> deposition/storage -> remobilisation

Diagram 2 - slope force and water

rain -> infiltration -> weight + pore pressure -> lower effective strength
road/river cut -> toe support loss -----^ -> failure

Diagram 3 - aquifer section

recharge area -> unconfined water table -> aquitard -> confined aquifer
well rises to potentiometric level

Diagram 4 - coastal upconing

coast | freshwater wedge over saline water
| pumping well -> cone of depression -> saline rise/upconing

Map 5 - India recall

Himalaya/NE: landslides + springs | Western Ghats/Nilgiris: rain-weathered slopes
Chambal: ravines | Indo-Gangetic plain: alluvial aquifers/food
Peninsula: hard-rock aquifers | Coast: saline interface

Answer skeleton 6

definition -> labelled diagram/map -> mechanism -> India evidence
-> effect/evaluation -> limitation -> matched response -> qualified conclusion